

Onboarding SOP for Merchants, Banks & TSPs

The single procedure for bringing a merchant live on Katana, and for getting Katana itself onboarded by a sponsor bank, payment aggregator or TSP. Every step is split by rail (P2P, UPI Intent and Payout) and comes with its documents, APIs, go-live gates and turnaround time.

P2P

Collection into a business VPA, confirmed by on-device or email capture and UTR matching

UPI Intent

Signed server-to-server order, paid by intent deeplink, QR or hosted page

Payout

Disbursement to approved beneficiaries by bank, UPI, wallet or USDT under maker-checker

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Contents

FOUNDATIONS

1 Purpose, scope & TAT rules

2 The three rails, side by side

3 Roles & ownership

PART A: ONBOARDING A MERCHANT ONTO KATANA

4 Common stage path & stage gates

6 P2P onboarding

5 Document checklist

7 UPI Intent onboarding & API

8 Payout onboarding & API

PART B: KATANA ONBOARDING TO A BANK OR TSP

9 Partner types by rail

11 What Katana submits

10 Partner stage path & TAT

12 What to obtain from the partner, by rail

13 Technical integration & UAT matrix

CONTROL

14 Master TAT table

16 Open decisions before approval

15 Tracker, status format & escalation

A Appendix: API quick reference

1 Purpose, scope & TAT rules

This SOP governs onboarding in two directions. Support after go-live is governed by KTN-SOP-OPS-001.

DIRECTION	WHO IS ONBOARDED	WHO ISSUES CREDENTIALS	WHO CALLS WHOM	SECTION
A · Inbound (Katana provides)	A merchant, branch or banker using Katana to collect or pay out	Katana issues Key + Salt to the merchant	Merchant calls Katana; Katana calls merchant back	§4–§8
B · Outbound (Katana consumes)	Katana, as platform / TSP, onto a sponsor bank, PA or TSP	The partner issues MID, client ID and secret to Katana	Katana calls partner; partner calls Katana back	§9–§13

TAT RULES

- **TAT is counted in business days (BD)**, Mon–Sat excluding bank holidays, from the day a **complete** document set is received. Hours are stated where used.
- **The clock pauses** while Katana waits on the merchant or partner (missing document, UAT fix, unsigned agreement). Pauses are logged on the tracker with a reason.
- **Target** figures are internal commitments Katana controls. **Partner-dependent** figures are tracked and chased, not promised to anyone.
- A stage more than 1 BD past target is flagged in the Ops group. More than 3 BD past target goes to the Management group (§15).

API AVAILABILITY TODAY

RAIL	MERCHANT-CALLABLE API	WHAT A MERCHANT CAN DO TODAY
INTENT	LIVE	Full self-service server-to-server integration (§7).
P2P	LIVE order · TO BUILD collect	A merchant server can create orders on the same order API (§7.2) paying into its own business VPA; the credit is confirmed by Katana Agent / email capture. The separate UTR-record collect API in §6.5 is specified but not released.
PAYOUT	TO BUILD	Beneficiaries and payouts are raised in the Katana portal under a session login and executed by an operator desk. There is no bank payout adapter (IMPS / NEFT) in production yet. The Key + Salt payout API in §8.6 is specified but not released.

Never promise a TO BUILD API in a sales or onboarding call. Merchants may design against the spec, but no go-live date may depend on it until Tech confirms release in writing.

2 The three rails, side by side

Every request is tagged with one or more rails at intake. The rail decides the documents, integration path, controls, TAT and partner.

ATTRIBUTE	P2P	UPI INTENT	PAYOUT
What it is	Customer pays a UPI VPA directly; Katana captures the credit and matches it to the order	Merchant server creates a signed order; customer pays by intent deeplink, QR or hosted page	Funds move out to a pre-approved beneficiary

ATTRIBUTE	P2P	UPI INTENT	PAYOUT
Money lands / leaves	Merchant's own business VPA / bank account	Collection VPA pool or acquirer MID per merchant config	Leaves from the merchant's payable balance (collected – settled – in-flight)
Confirmation source	Agent (SMS / app push) or email capture → match ladder: UTR → amount + VPA + recency → amount + recency (30-min window, trusted source)	Credit confirmation → signed callback to merchant + status enquiry	UTR / tx hash recorded by operator, proof uploaded
Integration artefacts	Katana Agent APK, trusted device, merchant code, optional email ingest	Key + Salt, order API, callback URL, return URL	Limits, beneficiary whitelist, maker-checker users
Key controls	Business VPA only · per-txn / daily amount / daily count limits · duplicate-UTR block	Signature check · ₹50,000 high-value hold · final-status lock · callback egress guard	Maker ≠ checker · approved beneficiary · balance check · ≥ ₹50,000 second approval
Default risk tier	Medium	Low–Medium (by category)	High (enhanced due diligence)
Merchant TAT (target)	2–3 BD	3–5 BD	5–7 BD
Partner TAT (Katana → partner)	2–4 weeks	3–6 weeks	4–8 weeks

P2P compliance gate. NPCI rules do not allow personal VPAs to be used for commercial collection. Every P2P merchant collects on a **business / merchant VPA** (P2M or small-merchant category) in the merchant's own legal name. Compliance rejects any application that routes business collections through a personal VPA. Confirm per-transaction and monthly caps for the VPA category with the issuing bank at onboarding; never quote limits from memory.

3 Roles & ownership

STAGE	ACCOUNT MGR	COMPLIANCE / RISK	TECH INTEGRATION	OPERATIONS	FINANCE	MANAGEMENT
Lead, rail tagging & commercials	R	C	—	—	C	A
KYC / KYB & risk tier	C	R/A	—	—	—	I
Account provisioning	I	C	C	R/A	—	—
Credentials issue (Key + Salt)	I	—	R	A	—	—
Integration, UAT & pre-go-live review	I	—	R/A	C	—	—
Payout limits & beneficiary approval	—	R (checker)	—	C	A	I

STAGE	ACCOUNT MGR	COMPLIANCE / RISK	TECH INTEGRATION	OPERATIONS	FINANCE	MANAGEMENT
Go-live sign-off	C	C	C	R	C	A
Partner (bank / TSP) onboarding	R	R	R	C	C	A

R responsible · A accountable · C consulted · I informed. One named person per role per onboarding is recorded on the tracker.

4 Common stage path & stage gates

All rails pass through the same eight stages. The dashboard onboarding funnel mirrors stages 2–6:

APPLICATION → DOCS_PENDING → SCREENING → BANK_VERIFY → CONFIG → LIVE.

- 1 **Intake & rail tagging** Day 0 AM
Log legal name, brand, business model, category, website / app, expected monthly TPV, transaction count and average ticket. Tag **P2P** **INTENT** **PAYOUT**. Reserve merchant code (e.g. **UK-108**). Create the lead (Leads → New lead; status **APPLICATION**). A merchant login with a one-time password is auto-created.
Exit: lead exists, rails tagged, document checklist (§5) sent.
- 2 **Document collection** merchant-dependent · clock not started AM
Status **DOCS_PENDING**. Chase at D+2 and D+5. A lead idle 15 BD without documents is closed and can be reopened.
Exit: complete set received and logged; **TAT clock starts**.
- 3 **KYC / KYB & risk review** 1 BD · Payout 2 BD Compliance
Status **SCREENING**. Verify entity (MCA / GST), directors and UBO ≥ 10 %, prohibited-category screen, sanctions & PEP screen, website / app policy pages, adverse media. Assign risk tier Low / Medium / High; High requires Management approval.
Exit: KYC **APPROVED**, tier recorded. Rejection is written with reasons.
- 4 **Bank account & VPA verification** 0.5 BD Compliance + Ops
Status **BANK_VERIFY**. Penny-drop or bank letter proves the settlement account is in the entity's name. For P2P, every receiving VPA is checked as a business VPA in the entity name.
Exit: settlement account and every VPA verified.
- 5 **Commercials & provisioning** 0.5 BD · Payout 1 BD AM + Finance → Ops
Status **CONFIG**. MDR / fees, settlement cycle, reserve %, limits. Anything outside the rate card needs Management. Ops sets rails, provider / branch mapping, Sub-MID (**TRAFFIC** or **KYC_APPROVED** mode), collection VPA, limits, and enables settlement.
Exit: merchant configured; commercial terms signed.
- 6 **Integration & UAT** P2P 0.5–1 · Intent 1–3 · Payout 2–3 BD Tech Integration
Run the rail checklist (§6–§8). A witnessed live test for every enabled rail; evidence (IDs, screenshots, callback body) is attached to the tracker.
Exit: every rail go-live checklist green; pre-go-live review passed (Intent).
- 7 **Go-live sign-off** same day Ops (R) · Management (A)
Status **LIVE**. Ops confirms all gates; Finance co-signs Payout; Management signs High-tier merchants. Merchant added to its Telegram support groups (KTN-SOP-OPS-001 §10).
- 3 **Hypercare** 7 calendar days Ops + Finance
Daily reconciliation check, first settlement reviewed by Finance, every incident auto-raised one severity level. Close hypercare with a one-line note on the tracker.

5 Document checklist

DOCUMENT	PROPRIETOR	PARTNERSHIP / LLP	PVT LTD / LTD	RAILS
ENTITY & PEOPLE (ALL RAILS)				
PAN of entity	Individual PAN	✓	✓	All
Certificate of Incorporation / Partnership deed / LLP agreement	—	✓	✓	All
MOA & AOA	—	—	✓	All
GST certificate, or Udyam / Shop & Establishment	✓	✓	✓	All
Board resolution / authority letter naming the signatory	—	✓	✓	All
KYC of proprietor / partners / directors / UBO ≥ 10 % (PAN + masked Aadhaar)	✓	✓	✓	All
Registered and operating address proof	✓	✓	✓	All
Cancelled cheque or bank letter, account in entity name	✓	✓	✓	All
Signed merchant agreement & rate card acceptance	✓	✓	✓	All
RAIL - SPECIFIC				
Business VPA proof: merchant-app profile screenshot per receiving VPA	✓ every VPA that receives funds			P2P
Device list: phone model, owner, Device ID; mailbox for email capture	✓			P2P
Per-trader limits requested: per-txn, daily amount, daily count	✓			P2P
Website / app URL with T&C, refund, privacy and contact pages live	✓			INTENT
Callback URL (HTTPS, public), return URL, tech SPOC, signing scheme preference	✓			INTENT
Beneficiary list: name, account + IFSC / VPA / wallet + network	✓			PAYOUT
Purpose-of-payout note & source-of-funds declaration	✓			PAYOUT
Maker and checker users (two different named people) & requested limits	✓			PAYOUT
Last 6 months bank statement (enhanced due diligence)	✓			PAYOUT

Handling. Documents are uploaded to the merchant KYB record (Merchant → Documents; PDF, PNG, JPEG or WEBP, max 12 MB each, types PAN · GST · CIN · MOA · AOA · BOARD_RESOLUTION · BANK_STATEMENT · MCC_DECLARATION · OTHER). They are never kept in Telegram or personal email. Aadhaar is accepted only masked (first 8 digits hidden).

6 P2P onboarding P2P

API STATUS	INTEGRATION EFFORT	UAT STAGE TAT	END-TO-END TARGET
Capture live · collect API TO BUILD	Device setup, ~30 min per phone	0.5–1 BD	2–3 BD

No acquirer API sits on this rail. Payment confirmation comes from the phone or inbox that receives the money, so setup quality decides whether payments auto-close.

6.1 Prerequisites

- ☐ Merchant exists with a merchant code; every receiving business VPA registered (primary + backups).
- ☐ An Android phone logged into the business app (Paytm / PhonePe for Business, bank app) that receives the credit alerts. BharatPe is email-only: plan the email channel.
- ☐ Per-trader limits configured (defaults ₹1,00,000 per transaction · ₹10,00,000 per day · 500 transactions per day; change only with Risk approval).
- ☐ Supported on-screen RRN capture apps: Paytm for Business, PhonePe Business, Google Pay for Business. BharatPe and MobiKwik: SMS / email capture only.
- ☐ For bankers using QR switching: QR uploaded in Banker portal, approved by admin, assigned to a store (one active QR per store, one store per QR).
- ☐ Admin user available to trust devices.

6.2 Setup steps

#	STEP	OWNER	DETAIL
1	Install agent	Merchant (Tech guides)	Open https://katanapay.co/katana-agent.apk on the same phone that receives payment alerts. Play Protect may warn: More details → Install anyway.
2	Connection	Merchant	Base URL https://katanapay.co (unchanged) · unique Device ID (e.g. <code>uk108-phone1</code> ; never reuse across phones) · merchant code (must show ✓ verified) · Forward alerts ON → Save.
3	Permissions	Merchant	SMS permission · Notification access ON · Battery Unrestricted (the #1 cause of missed pushes). Screen-read / auto-capture only if RRN capture is in scope.
4	Test alert	Merchant	Tap Send test alert → Recent activity shows ✓ → UNMATCHED . "queued — failed to connect" is a network problem on the phone.
5	Trust device	Ops	Merchant → Transaction agent & permissions → Trust , or Transaction Intel → Devices. An untrusted (UNKNOWN) device can never auto-confirm.
6	Email channel	Ops + Tech	Recommended for business merchants; mandatory for BharatPe. Gmail with 2-Step Verification + App Password + IMAP; server env <code>EMAIL_INGEST_ENABLED=1</code> , <code>EMAIL_INGEST_USER</code> , <code>EMAIL_INGEST_PASSWORD</code> , <code>EMAIL_INGEST_MERCHANT</code> . One mailbox → one merchant. Polled every minute.

#	STEP	OWNER	DETAIL
7	Witnessed live test	Tech + merchant	Create an order with the merchant's real business VPA; pay a clean whole-rupee amount; order → SUCCESS and pay page shows "Payment received" within 2 min (push / SMS) or ~1 min poll (email).

6.3 Matching & order rules to brief the merchant on

- Matching ladder: **UTR exact → amount + payee VPA + recency → amount + recency**. A unique match within 30 minutes from a trusted source auto-confirms; anything ambiguous goes to the manual reconciliation queue (Transaction Intel → Manual cases).
- Orders expire after **15 minutes**; a real credit landing after expiry **auto-revives** the order to SUCCESS.
- Where two bankers share one VPA, the agent's merchant code is the only attribution key. A mistyped code orphans every capture.
- Use clean amounts. Do not ask customers to pay odd paise amounts as a matching trick.

6.4 Go-live gate

- ☐ Device card all four green: **Trusted · Notification access · Forwarding · Online**, for every phone
- ☐ Merchant code on every device matches the dashboard exactly; Device IDs unique
- ☐ Real test payment auto-confirmed inside 2 minutes; agent Recent activity shows a server outcome
- ☐ Email channel ingesting (where configured): poll reports `ingested ≥ 1` on the test
- ☐ Only business VPAs registered; limits set; merchant briefed on expiry, revival and manual queue

6.5 P2P collect API TO BUILD

The matching engine and its controls are live inside Katana. The merchant-callable signed endpoint below is the release specification. Share it only for design, never as a go-live dependency.

PURPOSE	METHOD	PROPOSED URL	AUTH
Record a collection	POST	/api/v1/p2p/collect	Key + hash (\$7.3)
Collection status	GET	/api/v1/p2p/collect/{id}	Key + hash

```
POST https://katanapay.co/api/v1/p2p/collect
{ "key": "mk_9f2c...", "txnid": "UK108-P2P-5501", "trader_id": "...",
  "amount": "2500", "expected_amount": "2500", "utr": "123456789012",
  "vpa": "payer@bank", "hash": "<signature>" }
```

MATCH RESULT	STATUS	HTTP	MEANING
CORRECT	SUCCESS	200	New UTR, amount matches. Credited.
WRONG_AMOUNT	MANUAL_REVIEW	200	Amount differs from <code>expected_amount</code> .
OVER_PER_TXN_LIMIT	MANUAL_REVIEW	200	Above trader per-transaction limit.
OVER_DAILY_AMOUNT	MANUAL_REVIEW	200	Above trader daily amount limit.
OVER_DAILY_COUNT	MANUAL_REVIEW	200	Above trader daily count limit.
DUPLICATE	FAILED	409	UTR already settled a collection somewhere on the platform. Investigate; never treat as success.

MATCH RESULT	STATUS	HTTP	MEANING
–	–	403 / 404	Trader inactive / unknown. Rejected collections are still written to the ledger.

Omitting `expected_amount` disables amount matching, so every collection evaluates as CORRECT. The released API must make it mandatory for merchants.

7 UPI Intent onboarding & API INTENT

API STATUS LIVE	BASE URL https://katanapay.co	UAT STAGE TAT 1–3 BD	END-TO-END TARGET 3–5 BD
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Server-to-server integration in any language; no SDK required. Merchant-facing material is white-labelled

Katana Pay. Never name the settling acquirer to a merchant. Share the public kit:

<https://katanapay.co/katana-pay-integration.html> and [Katana-API-Kit.pdf](#).

7.1 Onboarding steps

#	STEP	OWNER	TAT	DETAIL
1	Configure collection	Ops	0.25 BD	Set the merchant's collection UPI ID / VPA pool. Without it, orders return a valid-looking QR pointing at a default payee.
2	Issue Key + Salt	Tech	same day	Merchant → Integration → Generate Key + Salt. Salt shown once ; regenerating invalidates the old pair. Record the scheme: operator-issued accounts are commonly Scheme 1 (SHA-512); portal self-issued default to Scheme 2 (HMAC-SHA256, recommended). Send Key and Salt through separate channels ; never Telegram.
3	Callback & return URLs	Merchant → Ops	same day	HTTPS (policy; the API also accepts http, so Ops must check), publicly resolvable, not private / loopback / link-local. Saved on the merchant profile or passed per order. Ops also creates the merchant's webhook signing config (Admin → Webhooks); without it transport headers are signed with a placeholder secret. Merchants authenticate on body <code>HASH</code> only.
4	Merchant builds	Merchant	merchant-dependent	Order create (§7.2), signing (§7.3), callback verify (§7.4), status enquiry (§7.5), daily recon sweep (§7.7).
5	UAT	Tech	1–2 BD	Run the test matrix (§7.8) with small live amounts, or on sandbox if one is issued.
6	Pre-go-live review	Tech	0.5 BD	Katana verifies the merchant's handler: HASH verification, numeric amount + order check, idempotency, working recon sweep (§7.9).

7.2 Create order

POST `/api/v1/katana-pay/order` · JSON or form-encoded · idempotent on `txnid` · 201 new, 200 with `reused: true` if the `txnid` already exists.

FIELD	REQ	TYPE · LENGTH	DESCRIPTION
key	Yes	AN 8–64	Merchant Key (<code>mk_...</code>)

FIELD	REQ	TYPE · LENGTH	DESCRIPTION
txnid	Yes	AN 1–60	Merchant reference; idempotency key. Pattern <code>^[A-Za-z0-9_-]{1,60}\$</code> . Prefix with merchant code (UK108-...)
amount	Yes	NU 1–12	Rupees, e.g. <code>"100"</code> / <code>"12.50"</code> . No separators; > 0. Sign and send the identical string
hash	Yes	AN 64–128	Signature (§7.3)
productinfo	No	CH 1–1024	Order description; signed under both schemes
firstname	No	CH 1–150	Customer name; signed under Scheme 1 only
email	No	CH 6–120	Customer email; signed under both schemes
phone	No	NU 10	Without country code
customer_vpa	No	AN 3–120	Payer UPI ID, where known
mode	No	INTENT QR	Default <code>QR</code> . Use <code>INTENT</code> on mobile
receiver_vpa / receiver_vpas	No	AN / array ≤ 30	Override collection VPA; first used, rest backups
currency	No	CH 3	<code>INR</code> only
return_url · notify_url	No	CH 5–1024	http(s) only; notify_url must be publicly resolvable

POST `https://katanapay.co/api/v1/katana-pay/order`

```
{ "key": "mk_9f2c...", "txnid": "UK108-ESN78452", "amount": "100", "productinfo": "Order 1001",
  "firstname": "David", "email": "david@example.com", "phone": "7417456565", "mode": "INTENT",
  "return_url": "https://merchant.com/return", "notify_url": "https://merchant.com/callback",
  "hash": "<signature>" }
```

HTTP 201

```
{ "verified": true, "merchant": "UK-108", "reused": false,
  "order": { "id": "8d20c0b6-...", "order_id": "UK108-ESN78452", "amount": 100, "status": "PENDING" },
  "pay_url": "https://katanapay.co/pay/8d20c0b6-...",
  "upi_intent": "upi://pay?pa=...&am=100.00&tr=UK108-ESN78452",
  "deeplinks": { "upi": "upi://pay?..." }, "qr_payload": "upi://pay?..." }
```

Always check `reused` . If `reused: true` and `order.amount` differs from what was sent, the reference collided. Abandon it and do not show that `pay_url` . (Idempotency is being moved from platform-wide to per-merchant scope, migrations 0024/0025. Prefixing txnid with the merchant code is correct under both and stays mandatory.)

7.3 Signing the request

Scheme 2 · HMAC-SHA256 (recommended)

message = txnid|amount|productinfo|email

hash = HMAC_SHA256(key = KEY + SALT, message) → lowercase hex, 64 chars

Scheme 1 · SHA-512

seq = key|txnid|amount|productinfo|firstname|email|salt (11 pipes between email and salt)

hash = SHA512(seq) → lowercase hex, 128 chars

Omitted fields sign as empty strings in position. Avoid | ~ = in txnid and productinfo.

7.4 Callback to merchant

Exactly **one** callback per order at the first terminal status, POSTed to `notify_url` or the saved webhook. No callback is sent until Key + Salt exist.

```
{ "PAY_ID": "9323643809301818", "ORDER_ID": "UK108-ESN78452", "TXN_ID": "4694e5ce-...",  
  "AMOUNT": "100", "CURRENCY_CODE": "356", "STATUS": "Captured", "RESPONSE_CODE": "000",  
  "RRN": "123456789101", "RESPONSE_DATE_TIME": "2026-09-08T11:35:42.991Z", "HASH": "8EAC6B6D..." }
```

Verify: drop HASH → sort every remaining field name by byte order → join NAME=value with "~"
(empty values keep "NAME=") → append SALT → SHA-256 → UPPERCASE hex → constant-time compare

DELIVERY	RULE	HEADER	USE
Attempts	6: 1 m → 5 m → 15 m → 1 h → 6 h → 24 h	x-event-type	payin.status
Deadline	Reply 2xx within 8 s; slower counts as failed	x-attempt	> 1 means retry
		x-timestamp	Unix seconds at send
After 6 failures	Abandoned (~31 h). AM can replay on request	x-signature	Internal transport only. Merchants verify body HASH
Recommended reply	{ "STATUS": "TRUE" }		

7.5 Status enquiry

GET /api/pay-status/{order.id}, using Katana's UUID, not the txnid (a txnid returns 404).

Unauthenticated, unsigned; authoritative record. ?wait=1 long-polls ≤ 25 s: customer screens only, never batch sweeps.

```
{ "order_id": "UK108-ESN78452", "amount": 100, "currency_code": "INR", "status": "SUCCESS",  
  "terminal": true, "rrn": "123456789101", "mode": "INTENT", "proof_submitted": false }
```

7.6 Status & error codes

CALLBACK	CODE	ENQUIRY	ACTION	HTTP	MEANING · ACTION
Captured	000	SUCCESS	Fulfil	401	invalid key / signature mismatch : check scheme, amount string, pipe count
Pending	005	PENDING	Wait		

CALLBACK	CODE	ENQUIRY	ACTION	HTTP	MEANING · ACTION
Failed	004	FAILED	Do not fulfil	403	Merchant blocked
Expired	003	EXPIRED	Re-check before cancelling: late credit can succeed with no second callback	400	Invalid amount / validation error
				409	Update conflicts with final status
				5xx · timeout	Indeterminate: retry with same txnid, then enquire

Order lifetime: 15 min, then EXPIRED . **High-value hold:** amounts $\geq$ ₹50,000, or orders where the payer uploaded proof, stay PENDING without expiry until an operator decides.

7.7 Reconciliation the merchant must run

- Daily sweep of all non-final orders plus final ones from the last 7 days, via plain status enquiry; re-check every EXPIRED .
- Match Portal → Transactions → Export against the bank statement on RRN . Escalate unknown SUCCESS, fulfilled-but-not-SUCCESS, duplicate RRN, RRN missing at bank.

7.8 UAT test matrix

#	CASE	EXPECTED	EVIDENCE
I-1	Signed order, Intent mode on mobile	201, deeplink opens UPI app, payee shown is the merchant	Response body, screenshot
I-2	Successful payment	Callback Captured/000, HASH verifies, enquiry SUCCESS	Raw callback, merchant log
I-3	Failed / abandoned payment	Failed or Expired handled; no fulfilment	Merchant order state
I-4	Wrong hash	401 signature mismatch	Response
I-5	Same txnid twice	200, reused: true , same order.id	Both responses
I-6	Callback endpoint returns 500 once	Retry with x-attempt: 2 ; handler idempotent	Delivery log
I-7	Callback not received	Recon sweep closes order via enquiry	Sweep log
I-8	Amount with paise (e.g. 100.50)	Callback AMOUNT "100.5" compared numerically, passes	Merchant log

7.9 Go-live gate

- ☐ Production Key + Salt issued; Salt server-side only (not browser, app binary or Git); scheme confirmed
- ☐ Collection VPA verified on a real payment; txnid prefixed with merchant code; reused checked
- ☐ Callback handler verifies HASH, compares AMOUNT numerically, is idempotent on ORDER_ID, replies 2xx < 8 s
- ☐ Fulfilment only on server-side confirmation; browser-return parameters treated as display only
- ☐ Recon sweep scheduled; EXPIRED re-checked; UAT I-1 to I-8 passed with evidence on tracker

- ❑ **Merchant with its own acquirer MID:** their Key + Salt vaulted in Katana; webhook `https://katanapay.co/api/gateway/payu/webhook` (Payments · Successful + Failed · POST) added in their gateway dashboard; test callback shows `hash_verified: true`. Use the internal own-acquirer guide; never link it publicly.

8 Payout onboarding & API PAYOUT

API STATUS	MODES	UAT STAGE TAT	END-TO-END TARGET
Portal live · Key API TO BUILD	BANK · UPI · WALLET · USDT	2–3 BD	5–7 BD

No payout leaves without an approved beneficiary and sufficient payable balance. Above the high-value threshold, a second approver who is not the maker must also sign off.

Execution today is manual. Katana has no live bank payout adapter. An operator makes the transfer on the partner bank portal and records the UTR / tx hash in Katana. The payout-rail TAT and SLA therefore depend on desk staffing until a Part B payout partner is integrated. The order status callback (`x-signature` = HMAC-SHA256 over raw body) is sent once with **no retry**. Merchants must poll `GET /api/v1/orders/{id}/status`.

8.1 Onboarding steps

#	STEP	OWNER	TAT	DETAIL
1	Enhanced due diligence	Compliance	2 BD	Source of funds, purpose of payouts, 6-month statement, beneficiary profile review. Default tier High.
2	Set limits	Finance (A)	0.5 BD	Per-payout max, daily cap, high-value threshold (platform default ₹50,000), allowed modes and USDT networks.
3	Create maker & checker users	Ops	same day	Two different named people. The maker can never approve their own beneficiary.
4	Register beneficiaries	Merchant maker	merchant-dependent	Each created <code>PENDING</code> . Validate by penny drop where the partner supports it.
5	Approve beneficiaries	Compliance / checker	≤ 0.5 BD	Name match against KYC and penny-drop result. Only <code>APPROVED</code> beneficiaries are payable.
6	UAT	Tech + Finance	1–2 BD	One ₹1 payout per enabled mode to an internal beneficiary; one above-threshold payout through approval; ledger and balance movement verified.

8.2 Portal APIs in use today (session auth)

```
POST /api/v1/beneficiaries
{ "beneficiary_name":"Asha Traders", "bank_name":"HDFC", "account_number":"50100...", "ifsc":"HDFC0000123"
}
// or "upi_id", or "wallet_address" + "network"
→ 201 { "id":"...", "status":"PENDING" }

POST /api/v1/beneficiaries/{id}/approve { "decision":"approve" | "reject", "reason":"KYC matched" }

POST /api/v1/payouts
{ "beneficiary_id":"...", "amount":"25000.00", "currency":"INR", "settlement_mode":"BANK",
"purpose":"vendor settlement" }
→ 201 { "order":{ "order_ref":"...", "status":"QUEUED" | "HOLD", "approval_required":false } }
```

8.3 Payout lifecycle & operator desk

Below threshold → **QUEUED** to the FIFO operator queue (assign ≤ 2 min, accept ≤ 3 min). At or above threshold → **HOLD** pending second approval. Operator accepts → processes → completes with UTR / tx hash and proof, or rejects / holds with reason.

```
Settlement (KTN-SET-000001):
REQUESTED → ACCEPTED → PROCESSING → PAID | USDT_TRANSFERRED → UTR_SUBMITTED → VERIFIED → RECONCILED
exceptions: ON_HOLD · CORRECTION_REQUIRED · INVALID_BENEFICIARY · INSUFFICIENT_BALANCE · ESCALATED ·
FAILED · REVERSED
```

8.4 Errors

HTTP	MEANING	FIX
404	Beneficiary not found for this merchant	Beneficiary must belong to the merchant
409	Beneficiary not approved	Get checker approval
409	Insufficient payable balance	Wait for settlement of collections or reduce amount
409	USDT wallet missing / network not allowed	Update beneficiary or allowed networks

8.5 Go-live gate

- ☐ Limits, threshold, modes and networks recorded and signed by Finance
- ☐ Maker and checker are different people; every live beneficiary **APPROVED** with penny-drop or equivalent evidence
- ☐ ₹1 payout per mode completed with UTR / tx hash; above-threshold payout went through HOLD → approval
- ☐ Payable balance formula confirmed with merchant: collected – settled – in-flight
- ☐ Finance co-signs go-live; first live payout batch reviewed by Finance same day

8.6 Merchant payout API **TO BUILD**

PURPOSE	METHOD	PROPOSED URL
Register beneficiary	POST	/api/v1/payouts/beneficiaries
Create payout	POST	/api/v1/payouts/create
Payout status	GET	/api/v1/payouts/{id}

```
{ "key": "mk_9f2c...", "txnid": "UK108-P0-9001", "beneficiary_id": "...", "amount": "5000",  
  "currency": "INR", "settlement_mode": "BANK", "purpose": "vendor settlement", "hash": "<signature>" }
```

Release blockers: Key + Salt auth on the route, `txnid` idempotency, and a signed payout status callback. Until idempotency ships, a retried request creates a **second payout**. No merchant may build automatic retries against payouts.

9 Partner types by rail

Katana is the applicant. Pick the partner type per rail first, because it fixes the document pack, the diligence depth and the TAT.

RAIL	PARTNER TYPE	WHAT KATANA GETS	TYPICAL MODEL
P2P	Sponsor / issuing bank; merchant-app provider (Paytm, PhonePe for Business, BharatPe)	Business VPAs on a bank handle, credit alerts (SMS / push / email), statement / MIS access	Merchant holds VPA in own name; Katana is the capture and matching layer
INTENT	Payment aggregator (PA) / acquirer, or sponsor bank via a TSP	MID / Sub-MIDs, intent + dynamic QR API, S2S status webhook, settlement file	Katana as TSP / platform under the PA's licence, or merchant's own MID (Direction A)
PAYOUT	Bank with API banking (current / escrow / nodal account), or payout TSP	IMPS / NEFT / RTGS / UPI transfer API, beneficiary validation, balance, status, returns	Katana-operated account with maker-checker, or merchant's own account via TSP

10 Partner stage path & TAT

- Qualification & shortlist** 5 BD Management + AM
 Shortlist ≥ 2 partners per rail. Collect onboarding form, rate card, NDA and API documentation. Score on pricing, API maturity, uptime, settlement cycle and escalation access. Log on partner tracker.
Exit: NDA signed, API docs in hand, partner chosen.
- Corporate & compliance pack** 3–5 BD to assemble Compliance
 Submit the Katana pack (§11) complete and in the partner's format. Most partner delays originate here.
Exit: partner acknowledges complete pack in writing.
- Partner due diligence** partner-dependent · 1–3 weeks Compliance + Tech
 Video KYC of signatories, business-model call, InfoSec questionnaire and review (Tech attends), site / infra questionnaire, possible on-site audit for payout.
Exit: risk / credit approval letter or email.
- Commercial & legal agreement** 1–2 weeks Finance + Legal · Management signs
 MDR / per-transaction pricing, settlement cycle, reserve / security deposit, chargeback and fraud liability, data-sharing, SLA and escalation matrix, termination.
Exit: executed agreement.
- UAT integration** 1–2 weeks Tech
 Sandbox credentials, IP whitelisting both ways, signature / encryption scheme, callback wiring, recon file format. Run the partner's test cases and the §13 matrix; share evidence.
Exit: partner UAT sign-off.

5

Production go-live 3–5 BD after UAT sign-off Tech + Ops + Finance

Production keys vaulted (never in committed `.env`), production IPs whitelisted, live low-value transaction per rail, recon against partner report, first settlement credit verified by Finance, contacts added to escalation matrix.

Exit: partner marked LIVE on tracker (criteria in §13.4).

11 What Katana submits

GROUP	ITEMS	RAILS
Corporate	COI, MOA / AOA, PAN, GST, board resolution naming signatories, director & UBO KYC, registered-office proof, audited financials (2 years) or CA-certified net worth, 6-month bank statements	All
Regulatory	RBI PA / PA-CB licence or in-principle status, or partner-sponsored model declaration; AML / KYC policy; grievance redressal policy with nodal officer; data-localisation declaration; DPDP privacy notice	All
Security	CERT-In empanelled VAPT report with closure evidence per finding; network architecture & data-flow diagrams; encryption & key-management note; access-control policy; incident response plan; BCP / DR plan; PCI DSS AoC only where card data is in scope	All (Intent & Payout reviewed hardest)
Business	Business model note per rail; merchant category list and prohibited-category policy; projected volumes (monthly TPV, count, average ticket); merchant onboarding & risk policy (this SOP)	All
Rail-specific	P2P: VPA usage declaration (business VPAs only), capture-device security note · Intent: website / app list, refund & chargeback process · Payout: fund-flow diagram, beneficiary validation policy, maker-checker matrix, sanctions screening for beneficiaries	P2P INTENT PAYOUT
Technical	Static egress IPs (UAT + prod), callback URLs, SPOC list (Tech / Ops / Finance / escalation), UAT results	All

12 What to obtain from the partner, by rail

Tech does not start UAT until every row for the rail is in hand. Missing API documentation pauses the partner clock.

ITEM	P2P	INTENT	PAYOUT
Product	Business / merchant VPAs on sponsor handle; credit alerts	MID / Sub-MIDs; intent & dynamic QR; S2S webhook	Current / escrow account; IMPS / NEFT / RTGS / UPI payout API
API documents required	VPA creation / mapping; credit alert format & sender IDs; statement / MIS download	Create order / intent; dynamic QR; status enquiry; refund; webhook payload & hash; settlement report spec	Beneficiary validation (penny drop); transfer; transfer status; balance enquiry; return / bounce webhook; statement
Credentials	Portal logins, alert sender IDs	Base URL, MID / Pay ID, client ID, key / secret or salt	Client ID, secret, debit account, corporate ID

ITEM	P2P	INTENT	PAYOUT
Signature / security	—	Signing scheme with a worked example ; IP whitelist	HMAC or AES / RSA payload encryption; mTLS certificate; IP whitelist; partner-portal maker-checker
Limits	Per-txn & monthly caps for the VPA category	Per-txn, daily, MCC restrictions; high-value rules	Per-txn per mode, daily, beneficiary cooling period, cut-off times
Recon	Daily statement	Daily settlement file (T+1)	Daily transfer report including returns
Callback into Katana	—	Vendor callbacks: <code>POST /api/vendors/{vendor}/callback</code> with <code>x-timestamp</code> + <code>x-signature</code> (secret <code>VENDOR_SECRET_<VENDOR></code>); PA pay-in: <code>/api/vendors/poolpay/callback</code> ; generic status: <code>POST /api/v1/webhooks/payment-status</code> ; own-acquirer PayU: <code>/api/gateway/payu/webhook</code> . Issue one per partner; never reuse a secret.	
Support	Escalation matrix (L1 / L2 / L3 with phone), status page, maintenance notice period, incident RCA commitment		
Target end-to-end	2–4 weeks	3–6 weeks	4–8 weeks

13 Technical integration & UAT matrix

13.1 Direction A field mapping

When Katana consumes a partner API, record these on the partner config before writing code. Katana signs outbound calls with the partner's secret and receives notifications on a callback URL issued at onboarding.

FIELD	SOURCE	STORED
Base URL (UAT + prod)	Partner API doc	Config service, per environment
MID / Pay ID / corporate ID	Partner onboarding email	Config service
Client ID, secret / salt, encryption keys, certificates	Partner, via secure channel	Vault only; never repo, <code>.env</code> commits or chat
Signing scheme + worked example	Partner API doc	Unit test fixture reproducing the example byte-for-byte
Callback URL + IP ranges	Katana issues / partner provides	nginx allowlist (no IP allowlisting exists in application code)

Where it is configured in Katana. A PA for UPI Intent is set once per provider: `PUT /api/providers/{id}/integrations` (Super Admin) with `base_url`, `pay_id`, `client_id`, `return_url`, `callback_url`, `env` (SANDBOX / PROD). The secret and API key go to the vault (AES-256-GCM). PROD cannot be enabled without base URL + secret, and the config applies to every branch mapped to that provider. A single merchant can override it via `PATCH /api/merchants/{id}/payment-config` (`poolpay.enabled`, `env`, `pay_id`, `settlement_vpa(s)` ≤ 20). An own PayU MID goes in via `POST /api/merchants/{id}/gateway-mid`.

Build flag. Katana signs with HMAC / SHA-256 / SHA-512 only. A partner that mandates RSA payload encryption, mTLS client certificates or a status-enquiry API not yet mapped needs a Tech build estimate **before** the agreement stage. Add it to the partner TAT at that point, not after UAT starts.

13.2 Partner UAT matrix

#	RAIL	CASE	PASS CRITERIA
P-1	P2P	Credit to business VPA	Alert captured (SMS / push / email), UTR parsed, order auto-confirmed
P-2	P2P	Duplicate UTR replay	Second record rejected DUPLICATE
P-3	P2P	Statement vs captures	Every statement credit matches one capture
T-1	INTENT	Create intent / QR, pay	Partner webhook signature verifies; order SUCCESS; merchant callback fires
T-2	INTENT	Failure, timeout, late credit	Failed / Expired recorded; late credit revives; reconciler picks up missed webhook
T-3	INTENT	Refund (where offered)	Refund status reaches terminal, reflected in recon
T-4	INTENT	Settlement file T+1	File totals equal Katana captured totals minus fees
Y-1	PAYOUT	Penny drop	Name returned and matched
Y-2	PAYOUT	Transfer per mode	UTR returned; status terminal; balance reduced
Y-3	PAYOUT	Invalid account / return	Return webhook processed; funds back to payable balance
Y-4	PAYOUT	Status timeout	Status enquiry used; no duplicate transfer issued
X-1	All	Bad signature / wrong IP	Rejected both directions

13.3 Security gates before production keys

- ☐ Latest CERT-In VAPT findings closed or risk-accepted in writing
- ☐ Production secrets in vault; rotation owner named; access limited to named engineers
- ☐ Egress IPs static and whitelisted at the partner; inbound partner IPs allowlisted in nginx
- ☐ AGENT_SIGNING_SECRET , VAULT_MASTER_KEY , FIFO_WEBHOOK_SECRET and VENDOR_SECRET_* set to production values; no sandbox bypasses enabled
- ☐ OpenTelemetry traces and alerting on partner calls and webhooks

13.4 A partner is LIVE only when

- ☐ Production keys vaulted and live low-value transaction completed on every contracted rail
- ☐ That transaction reconciled against the partner's own report
- ☐ Finance verified the first settlement credit (Intent) or first transfer debit (Payout)
- ☐ Partner escalation contacts in the KTN-SOP-OPS-001 escalation matrix

14 Master TAT table

14.1 Merchant onboarding (business days from complete documents)

STAGE	OWNER	P2P	INTENT	PAYOUT
Intake & checklist sent	AM	Day 0	Day 0	Day 0
KYC / KYB & risk	Compliance	1	1	2 (EDD)
Bank / VPA verification	Compliance + Ops	0.5	0.5	0.5
Commercials & provisioning	AM, Finance, Ops	0.5	0.5	1
Integration & UAT	Tech	0.5–1	1–3	2–3
Go-live sign-off	Ops, Mgmt	same day	same day	same day + Finance
End-to-end target		2–3 BD	3–5 BD	5–7 BD
External commitment (max)		3 BD	5 BD	7 BD
Hypercare	Ops + Finance	7 days	7 days	7 days

14.2 Katana onboarding to partner

STAGE	KATANA-CONTROLLED?	P2P	INTENT	PAYOUT
Qualification & shortlist	Yes	3 BD	5 BD	5 BD
Compliance pack	Yes	3 BD	5 BD	5 BD
Partner due diligence	No	1 wk	1–2 wk	2–3 wk
Agreement	Shared	0.5–1 wk	1–2 wk	1–2 wk
UAT	Shared	3–5 BD	1–2 wk	1–2 wk
Production go-live	Shared	2 BD	3–5 BD	3–5 BD
End-to-end target		2–4 weeks	3–6 weeks	4–8 weeks

Only the Katana-controlled rows are internal commitments. Partner rows are planning figures. Record the partner's own stated TAT on the tracker the day it is given, and chase against that.

15 Tracker, status format & escalation

TRACKER COLUMNS (ONE ROW PER ONBOARDING)

- Merchant / partner code, direction (A / B), rails
- Stage, stage start date, target date, pause days + reason
- Named owner per role (§3)
- Risk tier; KYC result; credentials issued (date, scheme; never values)
- UAT case IDs passed, evidence links
- Go-live date; hypercare close date

ESCALATION

DELAY PAST TARGET	ACTION
> 1 BD	AM posts status in Ops group tagging stage owner
> 3 BD	Raised in Management group with blocker and ETA
Partner > 5 BD past stated TAT	Management-to-partner call

DAILY STATUS POST (AM)

```
#ONBOARD UK-108
Direction: A (merchant)
Rails: INTENT + PAYOUT
Stage: 6 · Integration & UAT
Day: 4 of 7 BD target (0 paused)
Passed: I-1..I-5 · Y pending
Blocker: beneficiary approval
Owner: Compliance
ETA: 17 Sep
```

```
#PARTNER BANK-X
Direction: B (Katana → partner)
Rail: PAYOUT
Stage: 3 · Partner due diligence
Week: 2 of 2-3 wk (partner-stated)
Blocker: InfoSec questionnaire Q41
Owner: Tech
ETA: 22 Sep
```

16 Open decisions before approval

These values are not yet fixed. Do not quote plausible numbers to merchants or partners until each is closed.

ITEM	RAIL	STATUS	OWNER
Sandbox base URL and sandbox Key + Salt for merchants	Intent	DECISION	Tech Lead
Min / max transaction, daily & monthly volume, rate limits	All	DECISION	Risk + Finance
High-value review SLA (₹50,000 hold is deployment-wide, not per merchant)	Intent, Payout	DECISION	Head of Ops
Settlement cycle, cut-off, minimum, statement location	All	DECISION	Finance Controller
Refund process and turnaround (no refund API today)	Intent	DECISION	Finance + Ops
Release dates for P2P collect API and payout Key API	P2P, Payout	TO BUILD	Tech Lead
Apply migration 0025 (per-merchant txnid) and update API Kit §5.1	Intent	PENDING	Tech Lead
Bank payout adapter (IMPS / NEFT / UPI) and retrying payout callback	Payout	TO BUILD	Tech Lead
Merchant webhook signing config auto-created at credential issue (removes placeholder secret)	Intent	TO BUILD	Tech Lead
/api/pay redirect path: scope txnid idempotency to merchant (same class as 0024)	Intent	TO BUILD	Tech Lead
Named owners per role; standard rate card; shortlisted partners per rail	All	DECISION	Management

A Appendix: API quick reference

Base URL <https://katanapay.co>. Share only rows relevant to the recipient; internal rows never leave Katana.

RAIL	STATUS	METHOD	PATH	AUTH	AUDIENCE
INTENT	LIVE	POST	/api/v1/katana-pay/order	Key + hash	Merchant
INTENT	LIVE	GET	/pay/{order.id}	Public	Customer
INTENT P2P	LIVE	GET	/api/pay-status/{order.id}[?wait=1]	Public (UUID)	Merchant
INTENT	LIVE	POST	merchant_notify_url	Body HASH (SHA-256 + Salt)	Merchant
P2P	LIVE	APK	/katana-agent.apk	Trusted device	Merchant
P2P	TO BUILD	POST GET	/api/v1/p2p/collect[/id]	Key + hash	Merchant
PAYOUT	LIVE	POST GET	/api/v1/beneficiaries	Portal session	Merchant maker
PAYOUT	LIVE	POST	/api/v1/beneficiaries/{id}/approve	Checker session	Checker
PAYOUT	LIVE	POST	/api/v1/payouts	Portal session	Merchant maker
PAYOUT	LIVE	POST GET	/api/v1/settlements	Admin / Finance	Internal
PAYOUT	TO BUILD	POST GET	/api/v1/payouts/beneficiaries /create /id	Key + hash	Merchant
Partner	LIVE	POST	/api/v1/webhooks/payment-status	x-signature HMAC-SHA256	Bank / TSP
Partner	LIVE	POST	/api/gateway/payu/webhook	Merchant's vaulted PayU salt	Own-acquirer merchant

Shareable merchant documents: Katana Pay integration guide (HTML + PDF), Katana API Kit v1.1 PDF, Katana Agent Guide PDF. Internal only: own-acquirer webhook guide, partner credentials, this SOP.

Prepared by

Head of Operations
Date:

Reviewed by

Compliance · Tech Lead · Finance
Controller
Date:

Approved by

Management
Date:

Revision history: v1.0 (14 Sep 2026) initial issue for Management approval. Supersedes the onboarding sections (§4–§9) of KTN-SOP-OPS-001 v1.0; support, severity and SLA sections remain there.